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Vinay Ram Gazula

Graduate Student at NJIT

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Aspiring data scientist with a solid foundation in mechanical engineering, seeking a role as a Summer Intern in the field of data science, to leverage an unique blend of analytical, engineering and programming skills.

Education

Master of Science, *New Jersey Institute of Technology* — **Sep 2023 - May 2025** | NJ, USA
Data Science Computational Track — GPA: 3.83
Relevant Coursework: Applied Statistics, Introduction to Big Data, Data Managment Systems Design, Advanced Database Systems Design, Machine Learning and Data Analytics with R Program

Bachelor of Technology, *SRM University AP* — **Aug 2017 - May 2021** | AP, India
Mechanical Engineering — GPA: 3.52
Awarded with merit scholarship of 50% tuition fee for the entire bachelor's degree

Skills

Programming Languages	Python, R, Bash, C/C++, Java, Pig, Hadoop, MATLAB	
Database Languages	SQL, MySQL, SQLite, PL/SQL, Oracle, BigQuery, MongoDB	
Data Analytics	Spreadsheets, Pandas, Tidyverse, Matplotlib, Plotly, Tableau	
Tools	**Git**	GitHub
	Cloud	AWS EC2
	IDE / Text Editor	VS Code, RStudio, Sublime Text, KATE, Vim
	Documentation	MS Office, LaTeX, R Markdown
	Operating Systems	Linux, MacOS, Windows, UNIX
Libraries	NumPy, SymPy, Tensorflow, Keras, PyTorch, yfinance, Streamlit, ggplot	

Experience

Research Assistant — **Oct 2023 - Present**
Under Dr. Jason T. L. Wang (Professor, Computer Science, NJIT) — Newark, USA

- Integrated Explainable AI (xAI) tools and leveraged interpretability into machine learning and deep learning tools used for space weather research in collaboration with NSF / NASA (`https://nature.njit.edu/solardb/`)
- Incorporated xAI tools LIME, SHAP, PDP and ALE plots into a transformer-based framework "SolarFlareNet", used to forecast whether a $\geq M$-class flare occurs within $24, 48$ & 72 hours using SHARP parameters
- Submitted a poster paper for the 37th International FLAIRS Conference

Research Intern — **Sep 2021 - Feb 2022**
Under Dr. Satya P. Jammy (Associate Professor, Mechanical Department, SRM University AP) — Amaravati, India

- Worked on a Science Education and Research Board (SERB) govt. of India project titled "Wall Effects in Shock Wave Boundary Layer Interaction"
- Contributed in the development of solver code in python for gaseous reactions and viscous properties
- Validated the solver for different flow configurations like Single ramp, Double ramp, etc
- Published a conference paper in 26th National and 4th International ISHMT-ASTFE Heat and Mass Transfer Conference (link)

Projects

eComputer Store Database System | *MySQL, Python, Streamlit, Pandas* — **Sep 2023 - Dec 2023**
Data Managment Systems Design Course Project — GitHub Link

- Designed and developed a working database for an eComputerStore with all mentioned functionalities
- Followed a traditional approach for building database by designing an Enhanced Entity Relationship (EER) diagram, transforming the EER model to a Relational database schema and creating the database tables in MySQL using the Relational schema
- Developed a web application using Streamlit which connects to the MySQL server using a pipeline

MURPH | *Python, SymPy, NumPy, OpenMP, CUDA, OP2-API, CFD* — **Dec 2022 - Ongoing**
In collaboration with Dr. Satya P. Jammy (Associate Professor, Mechanical Department, SRM University AP) — GitHub Link

- A Multi-dimensional (2-D/3-D), Unstructured-mesh and Reactions based Parallel solver for Hyper-sonic flow regimes (MURPH)
- New design approach is used to build the solver code in python, so that the parallel code generation can be fully automated
- A 2D Ideal gas CFD solver is developed, capable of running parallel on multicore CPUs and GPUs using OpenMP and CUDA respectively
- Work is underway for integrating species reactions and development of a 3D solver

AlgoTrade API | *Python, yfinance, Pandas, Tensorflow, ks-api-client* — **May 2022 - Oct 2022**
Personal Project — GitHub Link

- Fully automated NSE Stock/Equity trading-bot for Kotak Securities (KOTAK TRADE API) with integrated back testing and ML algorithms for price and trend predictions
- The bot gathers any desired stock's historical price data, performs technical analysis using technical indicators and places orders to the stock broker using the broker API

Certifications

Google Data Analytics Professional Certificate — **Dec 2022 - Apr 2023**
Coursera — Certificate

Algorithmic Trading & Quantitative Analysis Using Python — **Mar 2023 - Apr 2023**
Udemy — Certificate